

An Explicit 13-Vertex Graph with No Edgeless Five-Vertex Vertex-Minor

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Abstract

For a graph G , let $[G]_{\text{LC}}$ be its local-complementation orbit and put $\beta(G) = \max_{H \in [G]_{\text{LC}}} \alpha(H)$. We give an explicit graph on exactly 13 vertices for which $\beta(G) = 4$: the Paley graph $P(13)$, whose graph6 string is `LlthgsL'mEkLkL`. The proof is an exact, reproducible finite computation. A labeled breadth-first traversal closes after 711,440 graphs, using all 13 local-complementation moves at every state and full 78-bit adjacency keys for duplicate handling. A separate verifier checks reachability, all 9,248,720 closure moves, and all 1,287 five-subsets of every orbit member. Thus no orbit member has an independent five-set; an independent four-set in one member gives equality. Consequently $P(13)$ has no edgeless graph on five vertices as a vertex-minor. Under the convention of Bae's vertex-minor Ramsey numbers, the witness gives $R_{\text{vm}}(5) \geq 14$.

1 Statement and context

All graphs here are finite, undirected, and simple. Local complementation at a vertex v , denoted τ_v , toggles every adjacency between two distinct neighbors of v and changes no other adjacency. The labeled orbit of G is

$$[G]_{\text{LC}} = \{\tau_{v_t} \cdots \tau_{v_1}(G) : t \geq 0\},$$

where the empty sequence is allowed. The independence number of a graph H is $\alpha(H)$, and

$$\beta(G) = \max_{H \in [G]_{\text{LC}}} \alpha(H).$$

Recent work on vertex-minor Ramsey numbers gives general bounds and related growth questions [1]. Bae proves $R_{\text{vm}}(5) \geq 13$ and conjectures $R_{\text{vm}}(5) = 15$ [2]. The result below supplies the intermediate lower-bound witness. Our literature comparison is deliberately bounded: targeted checks of those two primary records and searches for the notation, graph6 string, code, and explicit witness located no equivalent construction as of 9 September 2026. This is not a claim of priority or an exhaustive review.

Let G have vertex set $\mathbb{F}_{13}$. Distinct i, j are adjacent when

$$j - i \pmod{13} \in Q := \{1, 3, 4, 9, 10, 12\}. \quad (1)$$

Because $Q = -Q$, this defines a simple undirected graph; it is the Paley graph $P(13)$ and has 39 edges.

Theorem 1. *The graph $G = P(13)$ defined by (1), whose graph6 encoding is `LlthgsL'mEkLkL`, has a labeled local-complementation orbit of 711,440 members and satisfies*

$$\beta(G) = 4.$$

In particular, there exists a simple graph on exactly 13 vertices with $\beta(G) \leq 4$.

An edgeless five-vertex graph E_5 is a vertex-minor of G exactly when some member of $[G]_{\text{LC}}$ has an independent five-set. Indeed, deletion of a vertex distinct from v commutes with τ_v on the surviving vertices, so all deletions in a vertex-minor sequence can be moved to the end. Theorem 1 therefore also says that E_5 is not a vertex-minor of $P(13)$.

2 The exact orbit certificate

Fix the vertex order $0, 1, \dots, 12$ and index the 78 unordered vertex pairs. A labeled graph is stored losslessly as its 78-bit upper-triangular adjacency word, split into a 64-bit low word and a 14-bit high word. Local complementation at v is computed by toggling precisely the pair bits $\{i, j\}$ for which i and j are distinct neighbors of v .

Starting from the graph6 decoding of G , the generator performs a FIFO breadth-first search. For each dequeued state it applies τ_v for every $v \in \{0, \dots, 12\}$. Its duplicate table compares both parts of the full adjacency word, so a hash collision cannot identify distinct graphs. The queue empties with 711,440 distinct labeled states.

The resulting file `paley13_lc_orbit.bin` begins with the eight-byte magic `LC13ORB2` and a little-endian 64-bit record count. Each subsequent 15-byte record contains

$$(\text{low}_{64}, \text{high}_{16}, \text{parent}_{32}, \text{moves}).$$

The root has sentinel parent and move values. Every other record points to an earlier record and names the local-complementation vertex by which it was discovered. The file has 10,671,616 bytes and SHA-256 digest

$$6\text{ee}96\text{dec}3\text{a}2\text{e}4\text{c}2\text{dc}bd4424366\text{e}3499\text{b}611024\text{c}5911\text{b}8\text{f}20\text{c}7\text{d}54\text{dca}1\text{e}b73\text{d}02.$$

The following elementary observation explains why the finite checks certify the complete orbit rather than merely a large sample.

Lemma 2. *Let S be a finite set of labeled graphs. If $G \in S$, every member of S is obtained from G by a finite sequence of local complementations, and $\tau_v(H) \in S$ for every $H \in S$ and every vertex v , then $S = [G]_{\text{LC}}$.*

Proof. Reachability gives $S \subseteq [G]_{\text{LC}}$. Conversely, the root is in S and closure under every generator implies inductively that every finite local-complementation sequence starting at G remains in S . Hence $[G]_{\text{LC}} \subseteq S$. $\square$

For the serialized set S , a separate exact verifier checks that the first word is both the graph6 decoding and the adjacency relation (1), that all 711,440 words are distinct and in range, and that every nonroot word is the stated local complement of its earlier parent. Thus every listed word is reachable. It then independently recomputes all

$$711,440 \cdot 13 = 9,248,720$$

local complements and finds each result in S . Lemma 2 proves that S is exactly the full labeled orbit.

3 Excluding independent five-sets

For a five-subset T of the vertex set, let M_T be the 78-bit mask of all $\binom{5}{2}$ pairs contained in T . If A_H is the adjacency word of H , then T is independent in H if and only if

$$A_H \& M_T = 0.$$

The independent verifier exhausts all $\binom{13}{5} = 1,287$ choices of T for every one of the 711,440 orbit words. None of the

$$711,440 \cdot 1,287 = 915,623,280$$

exact mask intersections vanishes. Hence $\alpha(H) \leq 4$ for every $H \in [G]_{LC}$, and so $\beta(G) \leq 4$. In zero-based orbit record 834, the vertices $\{0, 1, 2, 3\}$ induce no edge. Thus one orbit member has independence number at least four, proving $\beta(G) = 4$ and completing the proof of Theorem 1.

This is an exact finite argument: graph states, moves, duplicate comparisons, and subset tests use integers only. No floating-point tolerance, random sampling, isomorphism heuristic, or assumed orbit size enters the conclusion.

4 Reproducibility and limitations

The supplementary material contains the generator `lc13_search.cpp`, the orbit certificate `paley13_lc_orbit.bin`, the independent verifier `verify_paley13.py`, and its machine-readable result `paley13_verification.json`. From the project directory, the decisive commands are

```
clang++ -std=c++20 -O3 -DNDEBUG evidence/lc13_search.cpp \
  -o evidence/lc13_search
./evidence/lc13_search certify LlhthgsL\mEkLkL \
  evidence/paley13_lc_orbit.bin
/Users/mac/4prove-or-disprove-math/.research-venv/bin/python \
  evidence/verify_paley13.py LlhthgsL\mEkLkL \
  evidence/paley13_lc_orbit.bin \
  evidence/paley13_verification.json
```

An independently written verifier, using direct neighbor-pair toggling rather than the generator’s clique-mask implementation, reproduced the same orbit size, closure count, subset count, four-set witness, and certificate digest.

The computation resolves the stated 13-vertex existence question; it does not classify all 13-vertex graphs or determine the exact value of $R_{vm}(5)$. The dated literature search cannot establish novelty.

For a deliberately narrow methodological comparison, exact finite searches in unrelated discrete-geometric models have likewise used integer certificates and independent verification [3], while an unrelated eight-vertex graph-algebra computation uses canonical graph enumeration, graph6 representatives, and a separate verifier [4]. Neither reference supports a vertex-minor claim or any part of the theorem above.

Computational and AI assistance. OpenAI Codex assisted with candidate search, implementation of the exact certificate and verification programs, literature screening, and manuscript preparation. The mathematical conclusion does not rely on generated prose or model confidence: it is supported by the explicit graph, the finite certificate, the exact checks described above, and a separate implementation audit.

References

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